

Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

Objective :

1. To develop an Embedded C program for adding an array of 16-bit numbers using the 8051 microcontroller.
2. To store the resulting 32-bit sum in the internal RAM of the microcontroller.

Code:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
```

```
while(1); // keep output stable
```

$$\}$$

Output:

OUTPUT:

